

SOS Markov Chains Mid-Term Report

Ananya Garg

June 20, 2026

1 Probability and Expected value

1.1 Basic Concepts

Probability measures uncertainty by assigning a number between 0 and 1 to events. A value of 0 means the event is impossible, while 1 means it is certain. The **sample space** Ω is the set of all possible outcomes, and an **event** A is any subset of Ω .

Example: For a die roll,

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

1.2 Axioms of Probability

All probability results follow from three basic rules:

1. $P(A) \geq 0$ for any event A
2. $P(\Omega) = 1$
3. If $A \cap B = \emptyset$, then

$$P(A \cup B) = P(A) + P(B)$$

From these, we obtain useful identities such as:

$$P(A^c) = 1 - P(A)$$

and

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

1.3 Conditional Probability

Conditional probability describes how probabilities change when we restrict attention to another event. It is defined as:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B) > 0$$

This leads directly to the multiplication rule:

$$P(A \cap B) = P(A | B)P(B)$$

More generally, for multiple events:

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2 | A_1) \dots P(A_n | A_1, \dots, A_{n-1})$$

This perspective is useful because many complex problems can be solved by conditioning on simpler intermediate events.

1.4 Independence

Two events A and B are independent if knowing one does not change the probability of the other. Formally,

$$P(A \cap B) = P(A)P(B)$$

Equivalently,

$$P(A | B) = P(A)$$

It is important not to confuse independence with disjointness. If two events are disjoint and have positive probability, they cannot be independent.

1.5 Law of Total Probability and Bayes' Theorem

Often, it is useful to break a problem into cases. If $\{B_1, \dots, B_n\}$ forms a partition of Ω , then:

$$P(A) = \sum_{i=1}^n P(A | B_i)P(B_i)$$

This idea naturally leads to Bayes' theorem, which allows us to reverse conditioning:

$$P(B_i | A) = \frac{P(A | B_i)P(B_i)}{\sum_j P(A | B_j)P(B_j)}$$

This is especially important in applications where we update beliefs after observing data.

1.6 Combinatorics

In many problems, probabilities are computed by counting outcomes.

$$n! = n(n-1) \cdots 1$$

If order matters:

$$P(n, k) = \frac{n!}{(n-k)!}$$

If order does not matter:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Care must be taken to avoid overcounting, especially when outcomes are not distinguishable.

1.7 Useful Techniques and Insights

Expectation can often be computed in stages using conditioning:

$$E[X] = E[E[X | Y]]$$

This idea is extremely useful in simplifying complex problems.

Indicator variables provide another powerful tool. For an event A , define:

$$I_A = \begin{cases} 1 & \text{if } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Then,

$$E[I_A] = P(A)$$

This allows probabilities to be handled using linearity of expectation.

In more complex settings, conditional independence may arise, where:

$$P(A | B, C) = P(A | C)$$

This concept is fundamental in probabilistic modeling and will reappear in Markov chains and time series models.

1.8 Expected Value

1.8.1 What is Expected Value?

Expected value is the long-run average outcome. If you repeat an experiment many times, the average result approaches the expected value.

Example: Roll a fair die many times:

$$E[X] = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

You never roll 3.5, but the average converges to it.

Expected value is a weighted average of all possible values.

1.8.2 Definition

For a discrete random variable:

$$E[X] = \sum_x x \cdot P(X = x)$$

For a continuous random variable:

$$E[X] = \int_{-\infty}^{\infty} x f(x) dx$$

Each value is weighted by how likely it is.

1.8.3 Properties of Expectation

Linearity of Expectation

$$E[X + Y] = E[X] + E[Y]$$

More generally:

$$E[aX + bY + c] = aE[X] + bE[Y] + c$$

This holds even if X and Y are dependent.

Expectation of a Constant

$$E[c] = c$$

Scaling

$$E[aX] = aE[X]$$

1.8.4 Expectation of Functions

$$E[g(X)] = \sum g(x)P(x) \quad (\text{discrete})$$

$$E[g(X)] = \int g(x)f(x) dx \quad (\text{continuous})$$

You do not need the distribution of $g(X)$ to compute this.

Example: If $X \sim \text{Uniform}\{1, 2, 3\}$:

$$E[X^2] = \frac{1^2 + 2^2 + 3^2}{3} = \frac{14}{3}$$

1.8.5 Indicator Variables

Define:

$$I_A = \begin{cases} 1 & \text{if event } A \text{ occurs} \\ 0 & \text{otherwise} \end{cases}$$

Then:

$$E[I_A] = P(A)$$

Example: Let X be the number of heads in 3 coin tosses. Define:

$$X = I_1 + I_2 + I_3$$

Then:

$$E[X] = E[I_1] + E[I_2] + E[I_3] = 3 \cdot \frac{1}{2} = \frac{3}{2}$$

1.8.6 Law of Iterated Expectation

$$E[X] = E[E[X | Y]]$$

Break expectation into steps:

1. Fix Y
2. Compute expectation given Y
3. Average over Y

1.8.7 Expectation and Independence

If X and Y are independent:

$$E[XY] = E[X]E[Y]$$

This is only true under independence.

1.8.8 Variance

$$\text{Var}(X) = E[(X - E[X])^2]$$

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

1.8.9 Covariance

$$\text{Cov}(X, Y) = E[XY] - E[X]E[Y]$$

Positive covariance means variables move together, negative means they move in opposite directions, and zero means no linear relationship.

1.9 Common Distributions

Several standard probability distributions arise frequently in applications.

1.9.1 Bernoulli Distribution

A Bernoulli random variable takes values:

$$X = \begin{cases} 1 & \text{with probability } p \\ 0 & \text{with probability } 1 - p \end{cases}$$

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

Example: Coin toss (success/failure)

1.9.2 Binomial Distribution

The Binomial distribution models the number of successes in n independent Bernoulli trials:

$$X \sim \text{Binomial}(n, p)$$

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

Example: Number of heads in n coin tosses

1.9.3 Poisson Distribution

The Poisson distribution models the number of events occurring in a fixed interval:

$$X \sim \text{Poisson}(\lambda)$$

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda$$

Example: Number of arrivals in a time interval

1.9.4 Uniform Distribution

A continuous random variable uniformly distributed over $[a, b]$ has density:

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Example: Random number drawn from an interval

1.9.5 Normal Distribution

The Normal distribution is one of the most important continuous distributions:

$$X \sim \mathcal{N}(\mu, \sigma^2)$$

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

Example: Measurement errors, natural phenomena

2 Random Variables

2.1 What is a Random Variable?

A random variable is a function:

$$X : \Omega \rightarrow \mathbb{R}$$

It assigns a real number to each outcome in the sample space. Convert results into numbers so that we can compute averages, analyze distributions, and apply algebra and calculus.

Example: For a coin toss with $\Omega = \{H, T\}$, define:

$$X = \begin{cases} 1 & H \\ 0 & T \end{cases}$$

A random variable is completely described by its distribution. If two random variables have the same distribution, they behave identically in a probabilistic sense.

2.2 Types of Random Variables

Discrete Random Variables Take countable values and are defined using a probability mass function (PMF). Examples includes the number of heads or number of arrivals.

Continuous Random Variables Take values over intervals and are defined using a probability density function (PDF). Examples include time, temperature, and height.

Mixed Random Variables Some variables have both discrete and continuous components, with point probabilities and a continuous density.

Degenerate Random Variable A random variable that takes a single value:

$$P(X = c) = 1$$

It behaves like a constant.

2.3 Probability Mass Function (PMF)

$$P(X = x)$$

$$P(X = x) \geq 0, \quad \sum_x P(X = x) = 1$$

The PMF shows how probability is distributed across discrete values.

2.4 Probability Density Function (PDF)

$$f(x)$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

For continuous variables, $P(X = x) = 0$ and probability comes from areas under the curve.

$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

2.5 Cumulative Distribution Function (CDF)

$$F(x) = P(X \leq x)$$

The CDF accumulates probability up to a point.

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1$$

For discrete variables, the CDF has jumps, and the size of the jump at x is:

$$P(X = x)$$

For continuous variables:

$$f(x) = \frac{d}{dx} F(x)$$

$$F(x) = \sum_{t \leq x} P(X = t) \quad (\text{discrete})$$

2.6 Transformations of Random Variables

If:

$$Y = g(X)$$

we determine the distribution of Y .

Transformations can stretch, compress, or reshape distributions.

$$P(Y = y) = \sum_{x: g(x)=y} P(X = x)$$

$$f_Y(y) = f_X(x) \left| \frac{dx}{dy} \right|$$

If multiple values of X map to the same Y , all must be included.

2.7 Joint Random Variables

We often consider multiple random variables:

$$X, Y$$

Their joint distribution is:

$$P(X = x, Y = y) \quad \text{or} \quad f(x, y)$$

This describes how variables behave together.

2.8 Marginal and Conditional Distributions

$$P(X = x) = \sum_y P(X = x, Y = y)$$

$$f_X(x) = \int f(x, y) dy$$

$$P(X = x | Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)}$$

The joint distribution gives the full picture, marginals ignore one variable, and conditional distributions fix one variable.

2.9 Independence

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

$$f(x, y) = f_X(x)f_Y(y)$$

Knowing one variable gives no information about the other.

Independence is stronger than zero covariance, which only indicates no linear relationship.

2.10 Symmetry

Some distributions are symmetric:

$$f(x) = f(2\mu - x)$$

This simplifies reasoning about the distribution.

3 Markov Chains

3.1 Introduction

Many real-world systems evolve over time under uncertainty. Examples include weather patterns, user behavior, and biological processes. In such systems, it is useful to model how the current state influences future outcomes.

Markov chains provide a framework for such systems, where the system evolves step by step and the future depends only on the present state.

3.2 Informal Idea

To build intuition, consider a simple example:

A restaurant serves only one item per day:

- Hamburger
- Pizza
- Hot dog

The choice for tomorrow depends only on what was served today.

This creates a system where states represent food items and transitions represent changes between days.

3.3 States

A state represents a possible condition of the system:

$$S = \{\text{Hamburger}, \text{Pizza}, \text{Hot dog}\}$$

- The set S is called the state space
- At any time, the system is in exactly one state

3.4 Transitions

A transition is the movement from one state to another in one time step.

- Each transition has an associated probability
- The system can move to a different state or remain in the same state

Mathematically:

$$P(X_{n+1} = j \mid X_n = i)$$

3.5 Transition Probabilities

These probabilities describe how the system evolves.

For each state:

$$\sum_j P_{ij} = 1$$

This means the system must transition to some state in the next step.

3.6 Memoryless (Markov) Property

The defining feature of a Markov chain is the memoryless property:

$$P(X_{n+1} \mid X_n, X_{n-1}, \dots) = P(X_{n+1} \mid X_n)$$

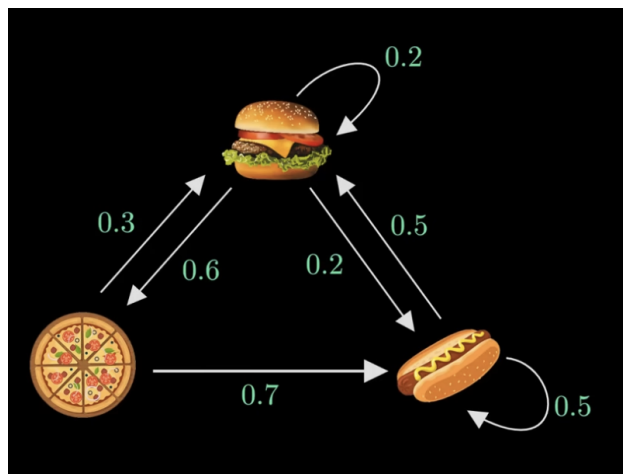
- Only the current state matters
- The past history is irrelevant

In simple terms, once the present is known, the past does not affect the future.

3.7 Graph Representation

A Markov chain can be visualized as a directed graph:

- Nodes represent states
- Edges represent transitions
- Edge weights represent probabilities



3.8 Formal Definition

A Markov chain is a stochastic process $\{X_n\}$ such that:

$$P(X_{n+1} = j \mid X_n = i, \dots) = P(X_{n+1} = j \mid X_n = i)$$

Define:

$$P_{ij} = P(X_{n+1} = j \mid X_n = i)$$

These are called transition probabilities.

3.9 Basic Properties of Markov Chains

Memoryless Property:

$$P(X_{n+1} \mid X_n, X_{n-1}, \dots) = P(X_{n+1} \mid X_n)$$

- The future depends only on the present state
- Past history does not influence future transitions

Time-Homogeneity:

$$P(X_{n+1} = j \mid X_n = i) = P_{ij}$$

- Transition probabilities do not change over time
- The system follows fixed rules

Row Sum Property:

$$\sum_j P_{ij} = 1$$

- From any state, total outgoing probability is 1
- The system must move to some state

Non-negativity:

$$P_{ij} \geq 0$$

- All probabilities are non-negative

3.10 Transition Matrix

The transition probabilities of a Markov chain can be represented compactly using a matrix:

$$P = [P_{ij}], \quad \text{where} \quad P_{ij} = P(X_{n+1} = j \mid X_n = i)$$

Structure and Interpretation

- Row i corresponds to the current state
- Column j corresponds to the next state
- Entry P_{ij} gives the probability of moving from state i to state j

Thus, each row of P represents a probability distribution over all possible next states.

Properties of the Transition Matrix

- $P_{ij} \geq 0$ for all i, j
- $\sum_j P_{ij} = 1$ for every state i (row-stochastic matrix)

Example

Consider a 3-state system:

$$P = \begin{bmatrix} 0 & 0.3 & 0.7 \\ 0.6 & 0.2 & 0.2 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

Interpretation

- From Pizza: 30% chance to go to Burger, 70% to Hotdog
- From Burger: 60% to Pizza, 20% stay, 20% to Hotdog
- From Hotdog: 50% stay, 50% to Burger

Why the Matrix Representation Matters

The transition matrix allows us to use linear algebra to study the evolution of the system. Instead of tracking paths individually, we can compute future distributions using matrix multiplication.

Connection to Linear Algebra

The transition matrix acts as a linear operator on probability vectors. If π_n is the distribution at time n , then:

$$\pi_{n+1} = \pi_n P$$

Each multiplication redistributes probability mass across states.

Intuition

Think of each row as describing how probability “flows” out of a state. Matrix multiplication aggregates all possible one-step transitions and spreads probability accordingly across the state space.

Important Insight

The entire behavior of a Markov chain is encoded in the matrix P . Once P and the initial distribution are known, all future behavior can be determined.

3.11 State Distribution and Evolution

At time n , the system is described by a probability distribution:

$$\pi_n = [P(X_n = i)]$$

The evolution of the system is given by:

$$\pi_{n+1} = \pi_n P$$

By repeated application:

$$\pi_n = \pi_0 P^n$$

Interpretation:

- Start with an initial distribution
- Apply the transition matrix repeatedly
- The distribution evolves over time

Intuition: Each multiplication spreads probability across states according to transition rules.

3.12 Example: Evolution of Distribution (Food Model)

Consider the food preference system with states:

$$S = \{\text{Pizza}, \text{Burger}, \text{Hotdog}\}$$

Let the transition matrix be:

$$P = \begin{bmatrix} 0 & 0.3 & 0.7 \\ 0.6 & 0.2 & 0.2 \\ 0 & 0.5 & 0.5 \end{bmatrix}$$

Suppose the initial distribution is:

$$\pi_0 = [1, 0, 0]$$

which means the system starts in the Pizza state with certainty.

After one step:

$$\pi_1 = \pi_0 P = [0, 0.3, 0.7]$$

After two steps:

$$\pi_2 = \pi_1 P = [0.18, 0.41, 0.41]$$

This example illustrates how the transition matrix redistributes probability over time.

3.13 n-step Transition Probabilities

The matrix P^n describes transitions over multiple steps:

$$(P^n)_{ij} = P(X_{t+n} = j \mid X_t = i)$$

Interpretation:

- Probability of going from state i to state j in n steps
- Accounts for all possible paths

Important insight:

$$(P^2)_{ij} = \sum_k P_{ik} P_{kj}$$

- Move from i to an intermediate state k
- Then from k to j
- Sum over all possible k

Key intuition: Matrix powers capture how probabilities propagate through the system.

3.14 Example: Weather Model

Consider a simple weather system with two states: [S = Sunny, Rainy]

Suppose the transition probabilities are given by:

$$P = \begin{bmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{bmatrix}$$

- If today is Sunny, tomorrow is Sunny with probability 0.8 and Rainy with probability 0.2
- If today is Rainy, tomorrow is Sunny with probability 0.4 and Rainy with probability 0.6

This system satisfies the Markov property since the weather tomorrow depends only on today's weather.

Such models are widely used in practice, including weather prediction, user behavior modeling, and financial market regimes.

3.15 Stationary Distribution

A distribution π is stationary if:

$$\pi P = \pi$$

- The distribution does not change over time
- The system is in equilibrium

3.16 Example: Finding the Stationary Distribution

Using the previous weather model, we compute the stationary distribution.

Let

$$\pi = (\pi_1, \pi_2)$$

where:

- π_1 = long-run probability of Sunny
- π_2 = long-run probability of Rainy

The stationary condition is:

$$\pi P = \pi$$

This gives the system of equations:

$$\pi_1 = 0.8\pi_1 + 0.4\pi_2, \quad \pi_2 = 0.2\pi_1 + 0.6\pi_2$$

Using $\pi_1 + \pi_2 = 1$, we solve:

$$\pi_1 = 0.8\pi_1 + 0.4(1 - \pi_1)$$

$$\pi_1 = 0.8\pi_1 + 0.4 - 0.4\pi_1$$

$$\pi_1 = 0.4\pi_1 + 0.4$$

$$0.6\pi_1 = 0.4 \quad \Rightarrow \quad \pi_1 = \frac{2}{3}$$

Thus,

$$\pi_2 = \frac{1}{3}$$

Final Answer:

$$\pi = \left(\frac{2}{3}, \frac{1}{3} \right)$$

3.17 State Classification

Understanding the long-term behavior of a Markov chain requires classifying its states.

Communicating States

State i communicates with state j if it is possible to go from i to j and from j to i in some number of steps.

If all states communicate with each other, the chain is called **irreducible**.

Transient and Recurrent States

A state i is called:

- **Recurrent** if the process returns to i with probability 1
- **Transient** if there is a non-zero probability that the process never returns to i

Intuition:

- Recurrent states are visited repeatedly over time
- Transient states may be visited only a finite number of times

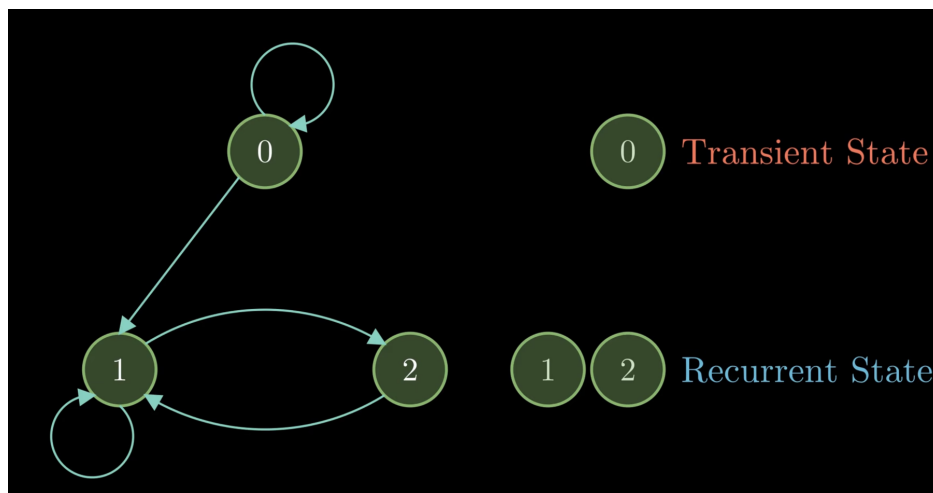


Figure 1: illustrating transient and recurrent states. State 0 is transient, while states 1 and 2 form a recurrent class.

Absorbing States

A state i is absorbing if:

$$P_{ii} = 1$$

Once the process enters this state, it never leaves.

Closed Classes

A set of states is called closed if the process cannot leave the set once it enters it.

These classifications help determine the long-term behavior of the system.

3.18 Chapman–Kolmogorov Equations

A key property of Markov chains is that multi-step transitions can be computed by combining shorter transitions.

For any $n, m \geq 0$:

$$P^{(n+m)} = P^{(n)} P^{(m)}$$

In particular:

$$P^n = \underbrace{P \cdot P \cdot \dots \cdot P}_{n \text{ times}}$$

Element-wise interpretation:

$$(P^2)_{ij} = \sum_k P_{ik} P_{kj}$$

- Move from state i to an intermediate state k

- Then from k to state j
- Sum over all possible intermediate states

Intuition

To go from i to j in multiple steps, the system must pass through intermediate states. These equations account for all such possible paths.

3.19 Ergodicity

A Markov chain is said to be **ergodic** if it satisfies:

- It is irreducible (all states communicate)
- It is aperiodic (no fixed cycle)

For an ergodic Markov chain:

- A unique stationary distribution exists
- The distribution π_n converges to π regardless of the initial state

Intuition

In an ergodic system, the process “forgets” its initial state over time and settles into a stable long-run behavior.

This property is crucial in applications such as simulation, statistical physics, and stochastic modeling.

3.20 Long-Term Behavior

The long-term behavior of a Markov chain depends on its structure.

- If the chain is irreducible and aperiodic, it converges to a unique stationary distribution
- If the chain has absorbing states, the system may eventually get trapped
- If the chain is reducible, different initial states may lead to different long-term outcomes

Convergence

For many Markov chains:

$$\lim_{n \rightarrow \infty} \pi_n = \pi$$

where π is the stationary distribution.

Interpretation

- Initially, the system depends on the starting state
- Over time, this dependence fades
- The system stabilizes into a predictable distribution

Understanding long-term behavior is one of the main goals of studying Markov chains.

4 Continuous-Time Processes and Poisson Models

4.1 Continuous-Time Markov Chains (CTMCs)

In a discrete-time Markov chain, transitions occur at fixed time steps. In many real-world systems, however, changes occur continuously over time. This motivates the study of **Continuous-Time Markov Chains (CTMCs)**.

A CTMC is a stochastic process $X(t)_{t \geq 0}$ that satisfies the Markov property:

$$P(X(t+s) = j \mid X(t), X(t_1), \dots) = P(X(t+s) = j \mid X(t))$$

- The future depends only on the present state
- Time is continuous rather than discrete

Key Difference from DTMC:

- DTMC: transitions happen at fixed steps
- CTMC: transitions occur at random times

4.2 Transition Rates and Q-Matrix

Instead of transition probabilities, CTMCs are described using **transition rates**.

Define the **generator matrix** (or Q-matrix):

$$Q = [q_{ij}]$$

where:

- $q_{ij} \geq 0$ for $i \neq j$ (rate of moving from i to j)
- $q_{ii} = -\sum_{j \neq i} q_{ij}$

Interpretation:

- q_{ij} represents how quickly transitions occur
- Rows of Q sum to zero

4.3 Poisson Process

The Poisson process is one of the most important continuous-time stochastic processes.

It models the number of events occurring randomly over time.

Let $N(t)$ denote the number of events up to time t . A Poisson process satisfies:

- $N(0) = 0$
- Independent increments
- Stationary increments

Distribution:

$$P(N(t) = k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

where $\lambda > 0$ is the rate of the process.

Examples:

- Number of calls arriving at a call center
- Number of trades in a financial market
- Arrival of customers in a queue

4.4 Exponential Waiting Times

A key property of the Poisson process is that the time between events follows an exponential distribution.

If T is the waiting time:

$$P(T > t) = e^{-\lambda t}$$

Memorylessness:

$$P(T > s + t \mid T > s) = P(T > t)$$

This means the process has no memory of how long it has already waited.

4.5 Memorylessness in Continuous Time

The exponential distribution is the only continuous distribution with the memoryless property.

This makes it fundamental in CTMCs, where the time spent in each state is typically exponentially distributed.

Intuition:

- The system does not “age” over time
- The probability of leaving a state is independent of how long it has been there

Connection to Markov Chains:

CTMCs can be viewed as an extension of discrete-time Markov chains where:

- Transitions still depend only on the current state
- But the timing of transitions is random and governed by exponential distributions